Jobsheet 2:

Database Operasional

Disusun Oleh:

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Jurusan Teknologi Informasi
[D4 Teknik Informatika/D4 Sistem Informasi Bisnis]
Politeknik Negeri Malang

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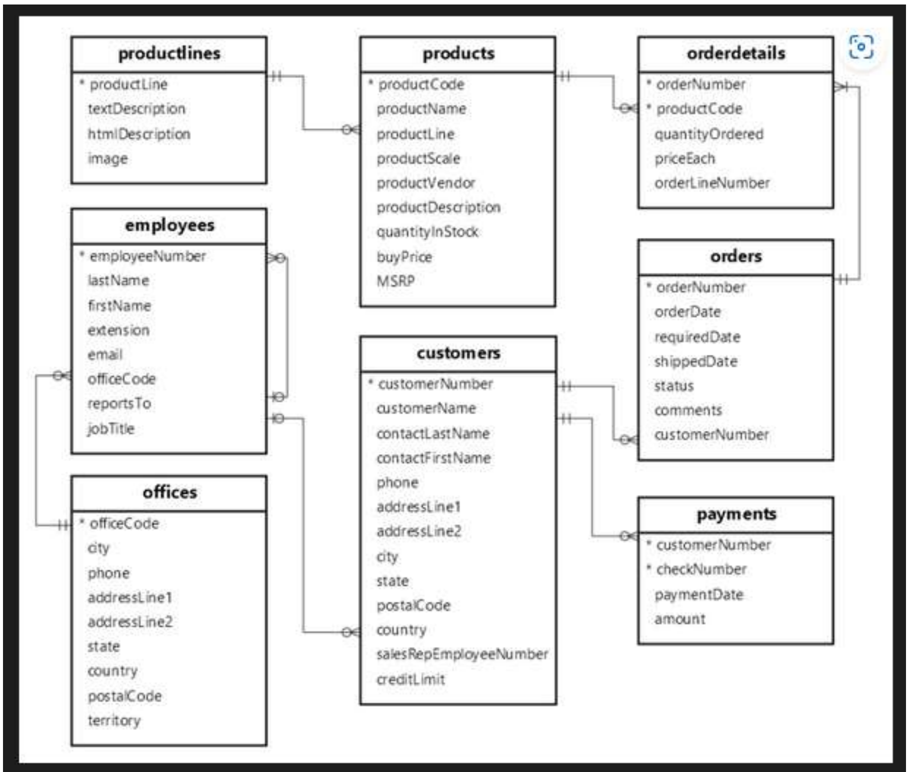
Tujuan Praktikum

Setelah melakukan praktikum ini, mahasiswa diharapakan dapat lebih mengenal data sumber, cara menganalisa serta melihat kebutuhan baik fungsional maupun non-fungsional dalam pengembangan data warehouse serta lebih memahami apa itu OLTP.

Studi Kasus :

LegendVehicle merupakan perusahan jual-beli tukar-tambah kendaraan klasik. Perusahaan ini memiliki cabang di berbagai negara. LegendVehicle memiliki sistem informasi ERP sendiri. Salah satu modul dari sistem ERP tersebut adalah modul penjualan. Desain database dari modul tersebut adalah sebagai berikut:



Gambar desain modul penjualan

Selain itu proses penjualan kendaraan pada perusahaan tersebut bukan hanya melalui showroom cabang, melainkan reseller-reseller bebas lainnya.

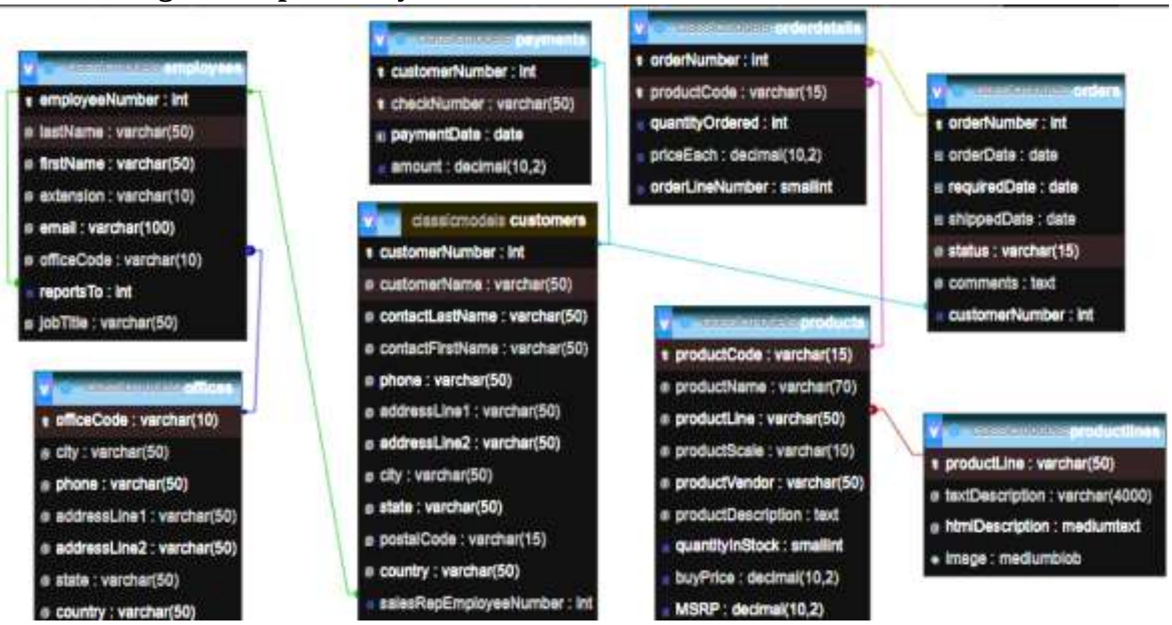
Data penjualan dari cabang tersebut dapat diunduh melalui link berikut: >> Data Penjualan <<

Tugas 1

1. **Import data perusahaan tersebut pada DBMS MySQL!**

	Table		Action						Rows	Type	Collation	Size	Overhead
	customers		Browse	Structure	Search	Insert	Empty	Drop	122	InnoDB	utf8mb4_0900_ai_ci	64.0 KiB	-
	employees		Browse	Structure	Search	Insert	Empty	Drop	23	InnoDB	utf8mb4_0900_ai_ci	48.0 KiB	-
	offices		Browse	Structure	Search	Insert	Empty	Drop	7	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
	orderdetails		Browse	Structure	Search	Insert	Empty	Drop	2,996	InnoDB	utf8mb4_0900_ai_ci	32.0 KiB	-
	orders		Browse	Structure	Search	Insert	Empty	Drop	326	InnoDB	utf8mb4_0900_ai_ci	32.0 KiB	-
	payments		Browse	Structure	Search	Insert	Empty	Drop	273	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
	productlines		Browse	Structure	Search	Insert	Empty	Drop	7	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
	products		Browse	Structure	Search	Insert	Empty	Drop	110	InnoDB	utf8mb4_0900_ai_ci	80.0 KiB	-
	8 tables		Sum						3,864	InnoDB	utf8mb4_0900_ai_ci	304.0 KiB	0 B

2. **Analisa struktur data dari database perusahaan tersebut, dalam bentuk tabel, analisa hubungan setiap tabel nya!**



Tabel 1	Tabel 2	Jenis Relasi
productlines	products	one to many
products	orderdetails	One to Many
orders	orderdetails	One to Many
customers	orders	One to Many
customers	payments	One to Many
employees	customers	One to Many
employees	employees	One to Many (Hierarchical)
employees	offices	Many to One

3. Analisa jumlah field pada setiap tabel!

Nama Tabel	Jumlah Field
productlines	4
products	9
orderdetails	5
orders	7

customers	13
payments	4
employees	8
offices	9

A. Analisa Data

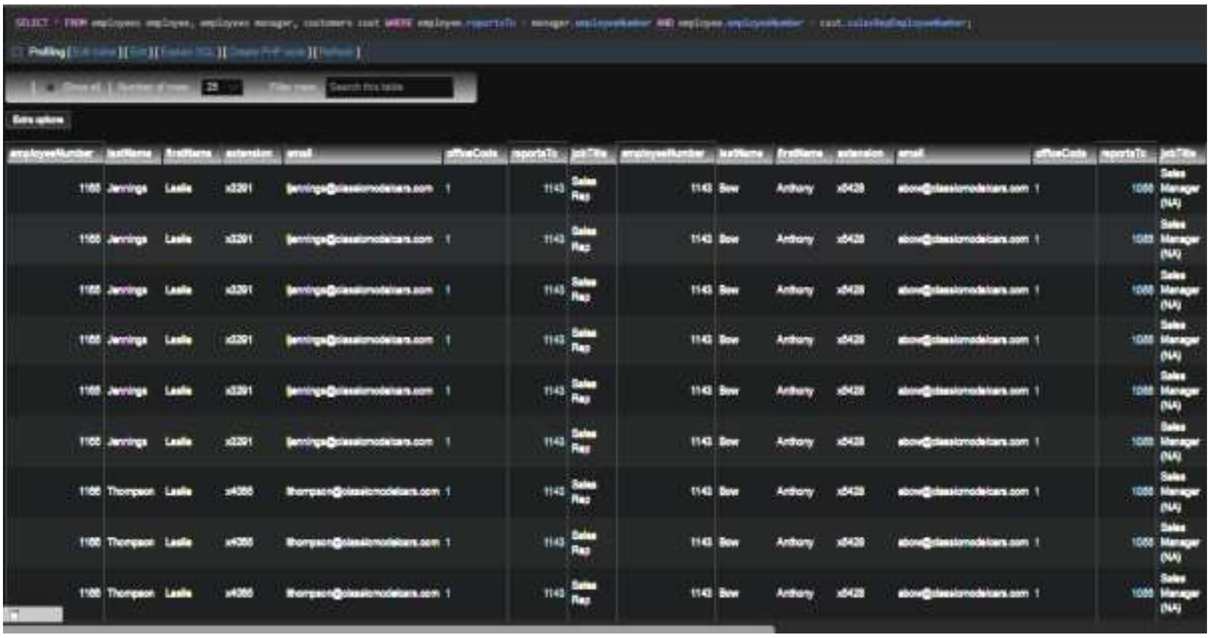
PERINGATAN: jika menemukan "ERROR" maka, Beranilah untuk menemukan dimana letak kesalahan untuk memberikan solusi. Jangan hanya bisa menyalahkan namun, tidak dapat memberikan solusinya.

PRAKTIKUM 1

1. Jalankan query berikut pada DBMS MySql yang telah tersedia data Perusahaan LegendVehicle.

   ```
   SELECT *
   FROM employees employe, employes manager, customer cust
   WHERE employee.reportsTo=manager.employeeNumber
   AND employee.employeeNumber=cust.salesRepEmployeeNumber;
   ```

   maka hasil dari query tersebut adalah data Employee beserta Manajernya dan Customer yang ia miliki. perhatikan hasil data dengan seksama.

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450	5677 Strong St.	NULL	San Rafael	CA	97562	USA	1165	210500.00
129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street	NULL	San Francisco	CA	94217	USA	1165	64600.00
161	Technics Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle	NULL	Burlingame	CA	94217	USA	1165	84600.00
321	Corporate Gift Ideas Co.	Brown	Julie	6505551386	7734 Strong St.	NULL	San Francisco	CA	94217	USA	1165	105000.00
450	The Sharp Gifts Warehouse	Frick	Sue	4085553659	3086 Ingle Ln.	NULL	San Jose	CA	94217	USA	1165	77600.00
487	Signal Collectibles Ltd.	Taylor	Sue	4155554312	2793 Furth Circle	NULL	Brisbane	CA	94217	USA	1165	60300.00
112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.	NULL	Las Vegas	NV	83030	USA	1166	71800.00
205	Toys4GrownUps.com	Young	Julie	6265557265	78934 Hillside Dr.	NULL	Pasadena	CA	90003	USA	1166	90700.00
219	Boards & Toys Co.	Young	Mary	3105552373	4097 Douglas Av.	NULL	Glendale	CA	92561	USA	1166	11000.00

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle	customerNumber	custome
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	205	Toys4Gro
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	219	Boards &
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	239	Collectab Designs I
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	347	Men 'R' L Ltd.
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	475	West Coa Collectab
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	173	Cambridg Collectab
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	204	Online M Collectab
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	320	Mini Crea
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	339	Classic G Inc.
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	379	Collectab Lass Inc.
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	486	Diecast C
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	187	Diecast C

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
205	Toys4GrownUps.com	Young	Julie	6265557265	78934 Hillside Dr.	NULL	Pasadena	CA	90003	USA	1166	90700.00
219	Boards & Toys Co.	Young	Mary	3105552373	4097 Douglas Av.	NULL	Glendale	CA	92561	USA	1166	11000.00
239	Collectable Mini Designs Co.	Thompson	Valarie	7605558146	361 Furth Circle	NULL	San Diego	CA	91217	USA	1166	105000.00
347	Men 'R' US Retailers, Ltd.	Chandler	Brian	2155554369	6047 Douglas Av.	NULL	Los Angeles	CA	91003	USA	1166	57700.00
475	West Coast Collectables Co.	Thompson	Steve	3105553722	3675 Furth Circle	NULL	Burbank	CA	94019	USA	1166	55400.00
173	Cambridge Collectables Co.	Tseng	Jerry	6175555555	4658 Baden Av.	NULL	Cambridge	MA	51247	USA	1188	43400.00
204	Online Mini Collectables	Barajas	Miguel	6175557555	7635 Spinnaker Dr.	NULL	Brickhaven	MA	58339	USA	1188	68700.00
320	Mini Creations Ltd.	Huang	Wing	5085559555	4575 Hillside Dr.	NULL	New Bedford	MA	50553	USA	1188	94500.00
339	Classic Gift Ideas, Inc	Cervantes	Francisca	2155554695	782 First Street	NULL	Philadelphia	PA	71270	USA	1188	81100.00
379	Collectables For Less Inc.	Nelson	Allen	6175558555	7825 Douglas Av.	NULL	Brickhaven	MA	58339	USA	1188	70700.00
486	Diecast Collectables	Franco	Valarie	6175552555	6251 Ingle Ln.	NULL	Boston	MA	51003	USA	1188	85100.00
187	Diecast Classics Inc.	Leong	Kelvin	2155551555	7586 Pompton St.	NULL	Allentown	PA	70267	USA	1216	100600.00

2. Buka tab baru pada browser untuk melakukan eksekusi query berikut:
   SELECT manager.employeeNumber as id_manager,
   CONCAT(manager.firstName," ,manager.lastName) as Manager,
   employee.employeeNumber as id_staff,
   CONCAT(employee.firstName," ",employee.lastName) as staff
   FROM employees employee, employees manager
   WHERE employee.reportsTo=manager.employeeNumber
   ORDER BY manager.firstName;

   dari hasil query diatas maka akan ditemukan atasan dari setiap pegawai.

id_manager	Manager	id_staff	Staff
1143	Anthony Bow	1165	Leslie Jennings
1143	Anthony Bow	1166	Leslie Thompson
1143	Anthony Bow	1188	Julie Firrelli
1143	Anthony Bow	1216	Steve Patterson
1143	Anthony Bow	1286	Foon Yue Tseng
1143	Anthony Bow	1323	George Vanauf
1002	Diane Murphy	1056	Mary Patterson
1002	Diane Murphy	1076	Jeff Firrelli
1102	Gerard Bondur	1337	Loui Bondur
1102	Gerard Bondur	1370	Gerard Hernandez
1102	Gerard Bondur	1401	Pamela Castillo
1102	Gerard Bondur	1501	Larry Bott
1102	Gerard Bondur	1504	Barry Jones
1102	Gerard Bondur	1702	Martin Gerard
1621	Mami Nishi	1625	Yoshimi Kato
1056	Mary Patterson	1088	William Patterson
1056	Mary Patterson	1102	Gerard Bondur
1056	Mary Patterson	1143	Anthony Bow
1056	Mary Patterson	1621	Mami Nishi
1088	William Patterson	1611	Andy Fixter
1088	William Patterson	1612	Peter Marsh
1088	William Patterson	1619	Tom King

TUGAS 2

1. Gambarlah hirarki organisasi berdasarkan atasan dari setiap pegawai sesuai dengan hasil prkatikum diatas!

SELECT
    manager.employeeNumber AS id_manager,

```
    CONCAT(manager.firstName, ' ', manager.lastName) AS manager_name,
    employee.employeeNumber AS id_staff,
    CONCAT(employee.firstName, ' ', employee.lastName) AS staff_name
FROM
    employees AS employee
JOIN
    employees AS manager
ON
    employee.reportsTo = manager.employeeNumber
ORDER BY
    manager_name;
```

id_manager	manager_name	id_staff	staff_name
1143	Anthony Bow	1165	Leslie Jennings
1143	Anthony Bow	1166	Leslie Thompson
1143	Anthony Bow	1188	Julie Firrelli
1143	Anthony Bow	1216	Steve Patterson
1143	Anthony Bow	1286	Foon Yue Tseng
1143	Anthony Bow	1323	George Vanauf
1002	Diane Murphy	1056	Mary Patterson
1002	Diane Murphy	1076	Jeff Firrelli
1102	Gerard Bondur	1337	Loui Bondur
1102	Gerard Bondur	1370	Gerard Hernandez
1102	Gerard Bondur	1401	Pamela Castillo
1102	Gerard Bondur	1501	Larry Bott
1102	Gerard Bondur	1504	Barry Jones
1102	Gerard Bondur	1702	Martin Gerard
1621	Mami Nishi	1625	Yoshimi Kato
1056	Mary Patterson	1088	William Patterson
1056	Mary Patterson	1102	Gerard Bondur
1056	Mary Patterson	1143	Anthony Bow
1056	Mary Patterson	1621	Mami Nishi
1088	William Patterson	1611	Andy Fixter
1088	William Patterson	1612	Peter Marsh
1088	William Patterson	1619	Tom King

2. Buka tab baru pada browser untuk melakukan eksekusi query berikut:

```
SELECT manager.employeeNumber as id_manager,
concat(manager.firstName," ",manager.lastName) as Manager,
employee.employeeNumber as id_staff, concat(employee.firstName,"
",employee.lastName) as staff,
count(cust.customerNumber) as total_cust
FROM employees employee join employees manager on
employee.reportsTomanager.employeeNumber
left join customers cust on
employee.employeeNumber=cust.salesRepEmployeeNumber
```

GROUP BY employee.employeeNumber
ORDER BY manager.firstName;
dari query tersebut menghasilkan jumlah customer dari setiap staff.

id_manager	Manager	id_staff	staff	total_cust
1143	Anthony Bow	1165	Leslie Jennings	6
1143	Anthony Bow	1166	Leslie Thompson	6
1143	Anthony Bow	1188	Julie Firrelli	6
1143	Anthony Bow	1216	Steve Patterson	6
1143	Anthony Bow	1286	Foon Yue Tseng	7
1143	Anthony Bow	1323	George Vanauf	8
1002	Diane Murphy	1056	Mary Patterson	0
1002	Diane Murphy	1076	Jeff Firrelli	0
1102	Gerard Bondur	1337	Loui Bondur	6
1102	Gerard Bondur	1370	Gerard Hernandez	7
1102	Gerard Bondur	1401	Pamela Castillo	10
1102	Gerard Bondur	1501	Larry Bott	8
1102	Gerard Bondur	1504	Barry Jones	9
1102	Gerard Bondur	1702	Martin Gerard	6
1621	Mami Nishi	1625	Yoshimi Kato	0
1056	Mary Patterson	1088	William Patterson	0
1056	Mary Patterson	1102	Gerard Bondur	0
1056	Mary Patterson	1143	Anthony Bow	0
1056	Mary Patterson	1621	Mami Nishi	5
1088	William Patterson	1611	Andy Fixter	5
1088	William Patterson	1612	Peter Marsh	5
1088	William Patterson	1619	Tom King	0

TUGAS 3

1. Siapakah staff dengan hirarki paling bawah yang berprestasi dilihat dari jumlah customer terbanyak?

Your SQL query has been executed successfully.

```
SELECT employee.employeeNumber AS id_staff, CONCAT(employee.firstName, " ", employee.lastName) AS staff, COUNT(cust.customerNumber) AS total_cust FROM employees AS employee LEFT JOIN customers AS cust ON employee.employeeNumber = cust.salesRepEmployeeNumber WHERE employee.employeeNumber NOT IN (SELECT DISTINCT reportsTo FROM employees WHERE reportsTo IS NOT NULL) GROUP BY employee.employeeNumber ORDER BY total_cust DESC LIMIT 1;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

id_staff	staff	total_cust
1401	Pamela Castillo	10

2. Jika KPI atasan dihitung dari customer yang dimilikinya dijumlah dengan customer dari staff dibawahnya, urutkan ranking prestasi keseluruhan pegawai beserta keterangan jumlah customer yang dimilikinya!

```
WITH RECURSIVE employee_hierarchy AS (
    -- Level awal: Semua pegawai dengan jumlah customer langsung mereka
    SELECT
        e.employeeNumber AS staff_id,
        e.firstName AS staff_firstname,
        e.lastName AS staff_lastname,
        e.reportsTo AS manager_id,
        COUNT(c.customerNumber) AS total_customers
    FROM
        employees e
    LEFT JOIN
        customers c ON e.employeeNumber = c.salesRepEmployeeNumber
    GROUP BY
        e.employeeNumber

    UNION ALL

    -- Rekursi untuk menjumlahkan customer dari bawahan ke atasan
    SELECT
        m.employeeNumber AS staff_id,
        m.firstName AS staff_firstname,
        m.lastName AS staff_lastname,
        m.reportsTo AS manager_id,
        eh.total_customers -- Menambahkan jumlah customer dari bawahan
    FROM
        employee_hierarchy eh
    JOIN
        employees m ON eh.manager_id = m.employeeNumber
)

-- Menghitung total KPI (customer langsung + bawahan)
SELECT
    eh.staff_id,
    CONCAT(eh.staff_firstname, ' ', eh.staff_lastname) AS staff_name,
    SUM(eh.total_customers) AS kpi_total_customers
FROM
    employee_hierarchy eh
GROUP BY
    eh.staff_id, staff_name
ORDER BY
    kpi_total_customers DESC;
```

staff_id	staff_name	kpi_total_customers
1002	Diane Murphy	100
1056	Mary Patterson	100
1102	Gerard Bondur	46
1143	Anthony Bow	39
1088	William Patterson	10
1401	Pamela Castillo	10
1504	Barry Jones	9
1323	George Vanauf	8
1501	Larry Bott	8
1286	Foon Yue Tseng	7
1370	Gerard Hernandez	7
1165	Leslie Jennings	6
1166	Leslie Thompson	6
1188	Julie Firrelli	6
1216	Steve Patterson	6
1337	Loui Bondur	6
1702	Martin Gerard	6
1611	Andy Fixter	5
1612	Peter Marsh	5
1621	Mami Nishi	5
1076	Jeff Firrelli	0
1619	Tom King	0
1625	Yoshimi Kato	0

3. Analisa kembali data LegendVehicle untuk mendapatkan ranking pegawai berdasarkan KPI "Jumlah omset yang didapat". Urutkan ranking pegawai beserta keterangan dana yang didapat!

```
WITH RECURSIVE employee_revenue AS (
    -- Level 1: Omzet langsung yang dihasilkan setiap pegawai dari customer yang mereka pegang
    SELECT
        e.employeeNumber AS staff_id,
        e.firstName AS staff_firstname,
        e.lastName AS staff_lastname,
        e.reportsTo AS manager_id,
        COALESCE(SUM(od.quantityOrdered * od.priceEach), 0) AS total_revenue
    FROM
        employees e
    LEFT JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
    LEFT JOIN orders o ON c.customerNumber = o.customerNumber
    LEFT JOIN orderdetails od ON o.orderNumber = od.orderNumber
```

```
    GROUP BY
        e.employeeNumber

    UNION ALL

    -- Rekursi: Menambahkan omzet bawahannya ke dalam total omzet atasannya
    SELECT
        m.employeeNumber AS staff_id,
        m.firstName AS staff_firstname,
        m.lastName AS staff_lastname,
        m.reportsTo AS manager_id,
        eh.total_revenue  -- Mewarisi omzet dari bawahannya
    FROM
        employee_revenue eh
    JOIN
        employees m ON eh.manager_id = m.employeeNumber
)

-- Hasil akhir: Ranking pegawai berdasarkan total omzet (langsung + bawahan)
SELECT
    eh.staff_id,
    CONCAT(eh.staff_firstname, ' ', eh.staff_lastname) AS staff_name,
    SUM(eh.total_revenue) AS kpi_total_revenue
FROM
    employee_revenue eh
GROUP BY
    eh.staff_id, staff_name
ORDER BY
    kpi_total_revenue DESC;
```

staff_id	staff_name	kpi_total_revenue
1002	Diane Murphy	9604190.61
1056	Mary Patterson	9604190.61
1102	Gerard Bondur	4520712.28
1143	Anthony Bow	3479191.91
1370	Gerard Hernandez	1258577.81
1088	William Patterson	1147176.35
1165	Leslie Jennings	1081530.54
1401	Pamela Castillo	868220.55
1501	Larry Bott	732096.79
1504	Barry Jones	704853.91
1323	George Vanauf	669377.05
1612	Peter Marsh	584593.76
1337	Loui Bondur	569485.75
1611	Andy Fixter	562582.59
1216	Steve Patterson	505875.42
1286	Foon Yue Tseng	488212.67
1621	Mami Nishi	457110.07
1702	Martin Gerard	387477.47
1188	Julie Firrelli	386663.20
1166	Leslie Thompson	347533.03
1076	Jeff Firrelli	0.00
1619	Tom King	0.00
1625	Yoshimi Kato	0.00

4. Jika KPI yang pertama merupakan "Jumlah customer yang bertransaksi" sedangkan KPI yang kedua "Jumlah omset yang didapat". Maka, berapakah jumlah field yang dibutuhkan untuk mendapatkan informasi tersebut?

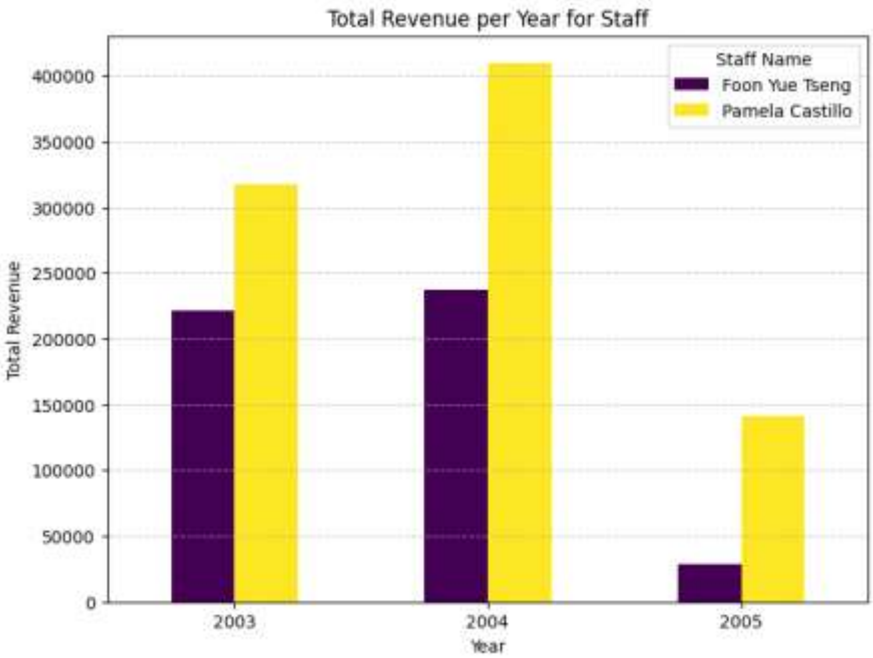
KPI	Number of Fields
Number of customers who transacted	3 fields (employeeNumber, EmployeeName, total_customer)
Total revenue obtained	4 fields (employeeNumber, employee_name, total_orders (handled by employee), total_revenue)

5. Buatlah report pertahun untuk KPI "Jumlah omset yang didapat" pada Foon Yue Tseng dan Pamela Castillo. Serta gambarkan grafiknya (grafik garis).

```
SELECT
    e.employeeNumber,
    CONCAT(e.firstName, ' ', e.lastName) AS employee_name,
    YEAR(o.orderDate) AS year,
    SUM(od.quantityOrdered * od.priceEach) AS total_revenue
FROM employees e
```

```
JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
JOIN orders o ON c.customerNumber = o.customerNumber
JOIN orderdetails od ON o.orderNumber = od.orderNumber
WHERE e.firstName IN ('Foon Yue', 'Pamela') AND e.lastName IN ('Tseng', 'Castillo')
GROUP BY e.employeeNumber, employee_name, year
ORDER BY year;
```

employeeNumber	employee_name	year ▲ 1	total_revenue
1286	Foon Yue Tseng	2003	221887.03
1401	Pamela Castillo	2003	317104.78
1286	Foon Yue Tseng	2004	237255.26
1401	Pamela Castillo	2004	409910.07
1286	Foon Yue Tseng	2005	29070.38
1401	Pamela Castillo	2005	141205.70



Studi Kasus

Pak Huhut merupakan pemegang saham LegendVehicle. dia membutuhkan dashboard untuk melihat perkembangan penjualan (omset) disetiap cabang di tiap tahunnya. Dikarenakan perusahaan tersebut belum merekrut Data Engineer maka, penarikan informasi hanya bisa dilakukan melaluai OLTP yang ada.

Analisalah terlebih dahulu:

1. Field apa saja yang diperlukan untuk menampilkan penjualan di setiap cabang.
   - branch_name (Nama cabang)
   - year (Tahun transaksi)
   - total_revenue (Total pendapatan/omset per cabang per tahun)

2. Bentuk query dengan memperhatikan relasi antar tabel.

```
SELECT
   o.city AS cabang,
   YEAR(od.orderDate) AS tahun_penjualan,
   SUM(odt.quantityOrdered * odt.priceEach) AS total_omset
FROM
   orders od
JOIN
   orderdetails odt ON od.orderNumber = odt.orderNumber
JOIN
   customers c ON od.customerNumber = c.customerNumber
JOIN
   employees e ON c.salesRepEmployeeNumber = e.employeeNumber
JOIN
   offices o ON e.officeCode = o.officeCode
GROUP BY
   o.city, YEAR(od.orderDate)
ORDER BY
   tahun_penjualan, total_omset DESC
LIMIT 0, 25;
```

cabang	tahun_penjualan ▲ 1	total_omset ▼ 2
Paris	2003	969959.90
London	2003	549551.94
San Francisco	2003	532681.13
NYC	2003	391175.53
Sydney	2003	304949.11
Boston	2003	301781.38
Tokyo	2003	267249.40
Paris	2004	1465229.84
London	2004	706014.52
NYC	2004	665317.99
Sydney	2004	542996.02
San Francisco	2004	517408.62
Boston	2004	457177.07
Tokyo	2004	151761.45
Paris	2005	648571.84
San Francisco	2005	376973.82
Sydney	2005	299231.22
London	2005	181384.24
Boston	2005	123580.17
NYC	2005	101096.20
Tokyo	2005	38099.22

SOAL BONUS: buatlah report lain dengan sumber data OLTP yang sama, analisa field yang digunakan, bentuk struktur query dan tuliskan dalam tabel serta grafiknya.

Answer :

equired Fields:

- p.productLine → Kategori produk dari tabel products.
- YEAR(o.orderDate) → Tahun transaksi dari tabel orders.
- od.quantityOrdered → Jumlah produk yang dipesan dari tabel orderdetails.
- od.priceEach → Harga per unit dari tabel orderdetails.
- p.productCode dan od.productCode → Kunci relasi antara tabel products dan orderdetails.
- od.orderNumber dan o.orderNumber → Kunci relasi antara tabel orderdetails dan orders.

```
SELECT
    p.productLine AS kategori_produk,
    YEAR(o.orderDate) AS tahun_penjualan,
    SUM(od.quantityOrdered * od.priceEach) AS total_omset
FROM
    orders o
JOIN
```

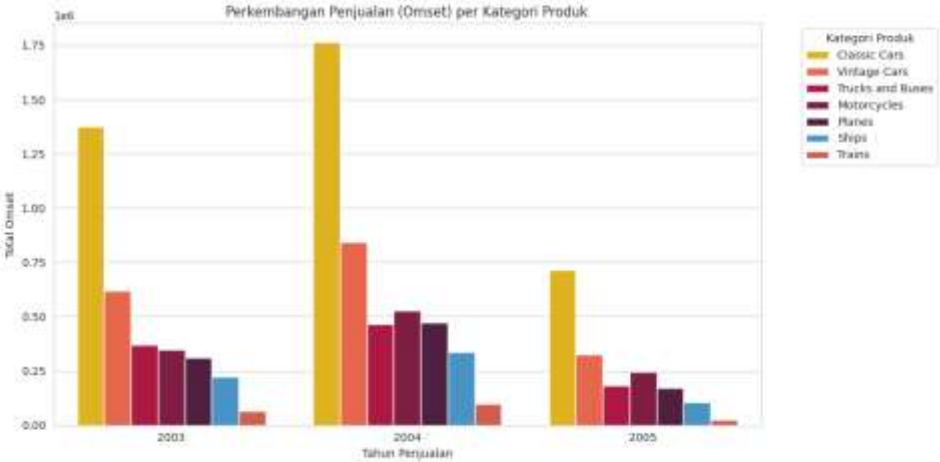
```
    orderdetails od ON o.orderNumber = od.orderNumber
JOIN
    products p ON od.productCode = p.productCode
GROUP BY
    p.productLine, YEAR(o.orderDate)
ORDER BY
    tahun_penjualan, total_omset DESC;
```

kategori_produk	tahun_penjualan	total_omset
Classic Cars	2003	1374832.22
Vintage Cars	2003	619161.48
Trucks and Buses	2003	370657.12
Motorcycles	2003	348909.24
Planes	2003	309784.20
Ships	2003	222182.08
Trains	2003	65822.05
Classic Cars	2004	1763136.73
Vintage Cars	2004	844581.85
Motorcycles	2004	527243.84
Planes	2004	471971.46
Trucks and Buses	2004	463990.00
Ships	2004	337326.10
Trains	2004	96285.53
Classic Cars	2005	715853.54
Vintage Cars	2005	323846.30
Motorcycles	2005	245273.04
Trucks and Buses	2005	182066.45
Planes	2005	172881.88
Ships	2005	104490.16
Trains	2005	26425.34

Kode :

```python
# Data dari gambar
data = {
    'kategori_produk': [
        'Classic Cars', 'Vintage Cars', 'Trucks and Buses', 'Motorcycles', 'Planes', 'Ships', 'Trains',
        'Classic Cars', 'Vintage Cars', 'Motorcycles', 'Planes', 'Trucks and Buses', 'Ships', 'Trains',
        'Classic Cars', 'Vintage Cars', 'Motorcycles', 'Trucks and Buses', 'Planes', 'Ships', 'Trains'
    ],
    'tahun_penjualan': [
        2003, 2003, 2003, 2003, 2003, 2003, 2003,
        2004, 2004, 2004, 2004, 2004, 2004, 2004,
        2005, 2005, 2005, 2005, 2005, 2005, 2005
    ],
    'total_omset': [
        1374832.22, 619161.48, 370657.12, 348909.24, 309784.20, 221182.08, 65822.05,
        1763136.73, 844581.85, 527243.84, 471911.46, 463990.08, 337326.10, 96285.53,
        715853.54, 323846.30, 245273.04, 182006.45, 172881.88, 104490.16, 26425.34
    ]
}

# Konversi ke DataFrame
df = pd.DataFrame(data)

# Menentukan warna untuk kategori produk
custom_colors = ["#FFC300", "#FF5733", "#C70039", "#900C3F", "#581845", "#3498DB", "#E74C3C"]

# Set tema dan palet warna
sns.set_style("whitegrid")
sns.set_palette(custom_colors)

# Visualisasi
plt.figure(figsize=(12, 6))
sns.barplot(x="tahun_penjualan", y="total_omset", hue="kategori_produk", data=df)

# Menyesuaikan judul dan label
plt.title("Perkembangan Penjualan (Omset) per Kategori Produk")
plt.xlabel("Tahun Penjualan")
plt.ylabel("Total Omset")

# Menyesuaikan posisi legenda agar lebih rapi
plt.legend(title="Kategori Produk", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()

# Menampilkan grafik
```

Hasil :



https://colab.research.google.com/drive/1-viLSeZ8WUEMh6-WXtPM2dmCCsjRkk0M?usp=sharing